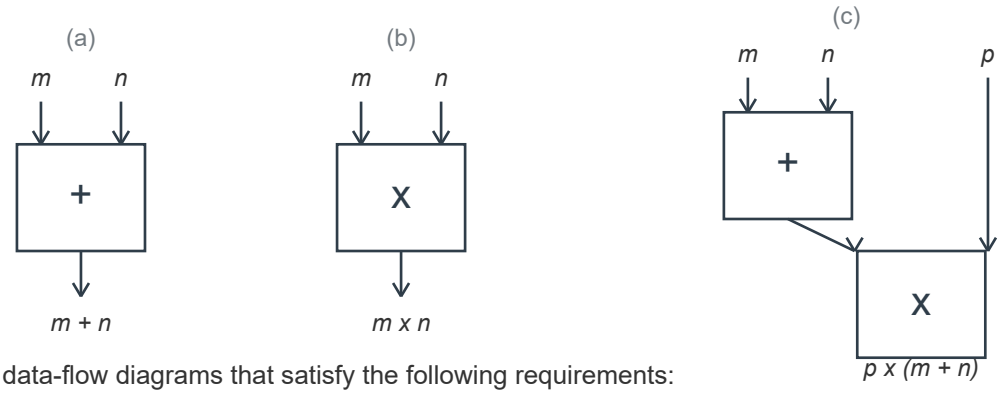
Homework 1

> Show the key steps in your reasoning. Unless a question states otherwise, bit positions are numbered from right to left starting at 0.

T1

We have two kinds of two-input black boxes. The first box adds its inputs, and the second box multiplies its inputs. An output may be connected to more than one later box, constants may be used as inputs, and an unlimited number of boxes is available.



Draw data-flow diagrams that satisfy the following requirements:

1. Compute F = (a + b)(c + d) + ab.
2. Using only MULT boxes, compute x^{15} with the minimum possible number of boxes.

T2

A system must assign a fixed-length binary code to each of 70 distinct symbols.

1. What is the minimum number of bits required for one symbol?
2. How many bit patterns are unused at that width?
3. How many bits are required for a string of N symbols?
4. If the symbol set grows to 129 symbols, how does the required width change?

T3

1. Convert +91 and -77 to eight-bit 2's complement binary numbers.
2. Interpret 01101011 and 10010110 as eight-bit 2's complement numbers and convert them to decimal.
3. Give the decimal range of a 12-bit 2's complement number. Then give the range of an N-bit 2's complement number.

T4

Let A = 101101 be a six-bit 2's complement number and B = 0010111 be a seven-bit 2's complement number.

1. Sign-extend both A and B to eight bits.
2. Compute A + B using eight-bit arithmetic.
3. Compute B - A using eight-bit arithmetic.
4. For each result, give both the binary representation and decimal value. Explain why filling every new high-order position with 0 would be incorrect in general.

T5

A fixed-width eight-bit ALU retains only the low eight bits of every sum. Analyze each addition twice: first as two 2's complement integers, and then as two unsigned integers.

```
01110110 + 00110101
```

```
11100110 + 11010101
```

1. Perform each addition and give the retained eight-bit result.
2. Give the decimal interpretation of both operands and the result under each data type.
3. State whether signed overflow and unsigned overflow occur in each case.
4. Summarize the condition that indicates signed overflow, and the condition that indicates unsigned overflow.

T6

All operands and results in this problem are exactly 16 bits. The NOT operation complements only those 16 bits. Express numerical results in hexadecimal.

1. (`x5A3C AND x0FF0`) OR `x8001`
2. NOT(`x00F0 XOR x0F0F`)
3. Use DeMorgan's Law to rewrite NOT(A OR B) using only AND and NOT.

T7

An eight-bit status vector represents devices 7 through 0. Bit i = 1 means device i is available, and bit i = 0 means it is busy. For each part, start from the original vector:

```
10111010
```

Give the mask, the operation, and the resulting vector for each task:

1. Use one AND operation to make devices 7 and 4 busy without changing other status bits.
2. Use one OR operation to make devices 0 and 2 available without changing other status bits.
3. Use one XOR operation to toggle the status of device 5.
4. Isolate the status of device 3 and move it into the most significant bit, with every other result bit equal to 0. You may use AND and repeated addition or a shift operation.

T8

Consider the following logical expressions:

- Q_1 = NOT((X AND Y) OR Z)
- Q_2 = (NOT X OR NOT Y) AND NOT Z

1. Complete the truth table.
2. State whether Q_1 and Q_2 are equivalent.
3. Use DeMorgan's Laws to verify your conclusion symbolically.

X	Y	Z	Q_1	Q_2
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

T9

Use the IEEE 754 single-precision floating-point format.

1. Convert `0 10000100 01101000000000000000000` to decimal. Show the sign, unbiased exponent, and significand.
2. Write the IEEE 754 single-precision representation of -13.625.
3. Give the bit patterns and power-of-two forms for the smallest positive normalized number and the smallest positive subnormal number.
4. Explain what is represented when the exponent field is all 1s and (a) the fraction field is all 0s or (b) the fraction field is nonzero.

Category	Exponent	Fraction	Meaning
Normalized	00000001 to 11111110	any	
Subnormal	00000000	nonzero	
Special	11111111	all 0s or nonzero	

T10

Use the following ASCII values:

Character	I	C	S	2	6	LF
Hex	49	43	53	32	36	0A

1. Write the string `ICS26\n` as a sequence of eight-bit binary ASCII codes.
2. Write the same byte sequence in hexadecimal.
3. Explain why uppercase and lowercase letters have different ASCII codes even when they share the same key on a keyboard.